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WEEK-7

Step – 1 Create a User — POST

```
Windows PowerShell x + - x
-----
dar--l    21-09-2026    14:48    controllers
dar--l    21-09-2026    14:48    node_modules
dar--l    21-09-2026    14:48    routes
-a--l     21-09-2026    14:44    39264 package-lock.json
-a--l     21-09-2026    14:44    349  package.json

PS C:\Users\saich\OneDrive\Desktop\HA\Practical week7>
nodemoo user-service> node app.js
User service running on port 5001
MongoDB connected

Windows PowerShell x + - x
PS C:\Users\saich\OneDrive\Desktop\HA\Practical week7>
odemeno\user-service> Invoke-RestMethod -Uri "http://localhost:5001/users"
-Method POST -ContentType "application/json" -Body '{
  "username":"chaitanya", "email":"Chaitanya27@gmail.com", "profile":{}
}'
message      user
-----
User created successfully @{username=chaitanya; email=Chaitanya27@gmail.com; ...}

PS C:\Users\saich\OneDrive\Desktop\HA\Practical week7>
odemeno\user-service> |
```

POST in Mongosh:

CHAITANYA's Workspace

Overview **POST** http://localhost:5001/users

Send

Body

```
1 {
2   "username": "Chaitanya",
3   "email": "Chaitanya27@gmail.com"
4 }
```

201 Created • 6 ms • 386 B

Body

```
1 {
2   "message": "User created successfully",
3   "user": {
4     "username": "Chaitanya",
5     "email": "Chaitanya27@gmail.com",
6     "_id": "K7mP3aF9zL6Qe2bD8rW1x",
7     "__v": 0
8   }
9 }
```

Step 2: Get User — GET

CHAITANYA's Workspace

Overview **GET** http://localhost:5001/u...

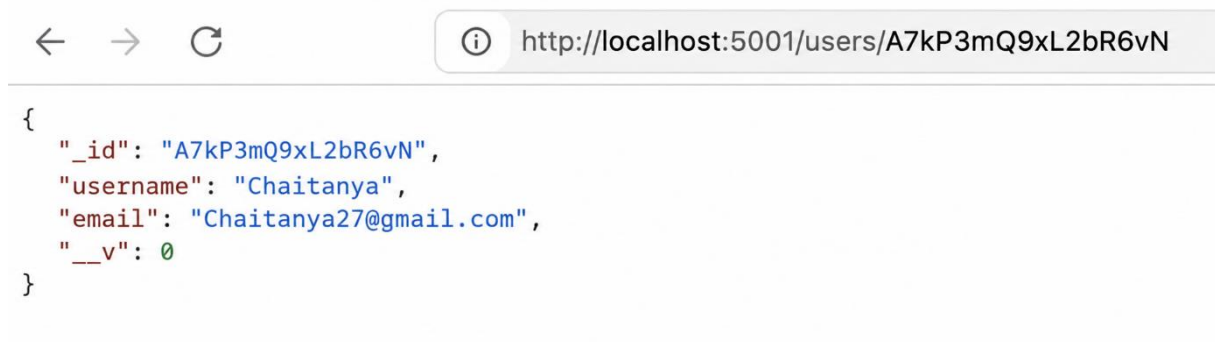
Send

Body

```
1 {
2   "_id": "68a9F2b1D7c4E9a6B3Xv0q1Z",
3   "username": "Chaitanya",
4   "email": "Chaitanya27@gmail.com",
5   "__v": 0
6 }
```

200 OK • 6 ms • 328 B

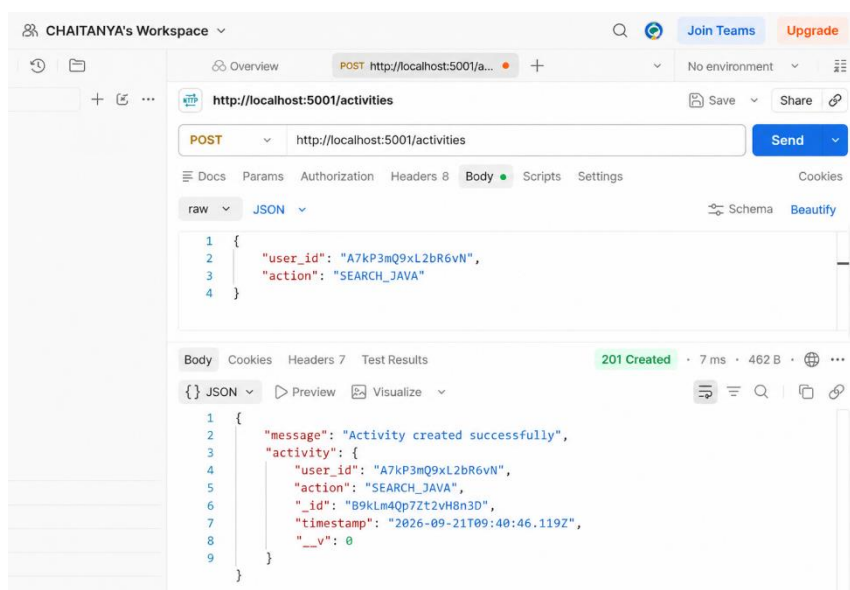
Output on browser by id:



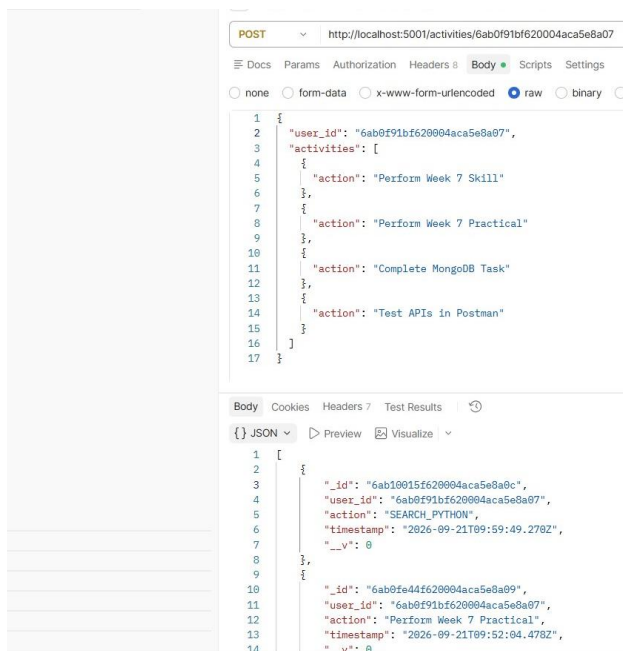
Step 3: Create Activity — POST



In mongosh:



Step 4: Add More Activities



```
PS C:\Users\saich\OneDrive\Desktop\HA\Practical week7> nodememo\user-service> Invoke-RestMethod -Uri "http://localhost:5001/activities/A7kP3mQ9xL2bR6vN" -Method GET

_id      : A7kP3mQ9xL2bR6vN
user_id  : A7kP3mQ9xL2bR6vN
action   : SEARCH_PYTHON
timestamp : 2026-09-21T09:59:49.270Z
__v      : 0

_id      : B9kLm4Qp7Zt2vH8n3D
user_id  : A7kP3mQ9xL2bR6vN
action   : Perform Week 7 Practical
timestamp : 2026-09-21T09:52:04.478Z
__v      : 0

_id      : C3dF8zL6mQ9vT2kR1X
user_id  : A7kP3mQ9xL2bR6vN
action   : SEARCH_JAVA
timestamp : 2026-09-21T09:40:46.119Z
__v      : 0
```

Output on browser:

```
Pretty print ☒

[
  {
    "_id": "6ab10015f620004aca5e8a0c",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "SEARCH_PYTHON",
    "timestamp": "2026-09-21T09:59:49.270Z",
    "__v": 0
  },
  {
    "_id": "6ab0fe44f620004aca5e8a09",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "Perform Week 7 Practical",
    "timestamp": "2026-09-21T09:52:04.478Z",
    "__v": 0
  },
  {
    "_id": "6ab0fb9ef620004aca5e8a08",
    "user_id": "6ab0f91bf620004aca5e8a07",
    "action": "SEARCH_JAVA",
    "timestamp": "2026-09-21T09:40:46.119Z",
    "__v": 0
  }
]
```

MongoDB connection:

MongoDB Compass - 127.0.0.1:27017/digital_library

Connections Edit View Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

- 127.0.0.1:27017
 - BookFlow
 - College
 - admin
 - config
 - digital_library
 - activities
 - users

127.0.0.1:27017 > digital_library

Open MongoDB shell Create collection Refresh

Collection name	Properties	Storage size	Data size	Documents	Avg. document size
activities	-	32.77 KB	300.00 B	3	100.00 B
users	-	36.86 KB	339.00 B	4	84.00 B

MongoDB Compass - 127.0.0.1:27017/digital_library.activities

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

- 127.0.0.1:27017
 - BookFlow
 - College
 - admin
 - config
 - digital_library
 - activities
 - users
 - local
 - student_records
- Dbmsl
- Week-3_Practical

127.0.0.1:27017 > digital_library > activities

Open MongoDB shell

Documents Aggregations Schema Indexes Validation

Type a query: { field: 'value' } or Generate query Explain Reset Find Options

25 1 - 3 of 3

```
{ "_id": ObjectId("6ab0fb9ef620004aca5e8a08"),
  "user_id": ObjectId("6ab0f91bf620004aca5e8a07"),
  "action": "SEARCH_JAVA",
  "timestamp": ISODate("2026-09-21T09:48:46.119+08:00") }
21_September_2026_at_09:48:46 UTC
__v: 0
```

```
{ "_id": ObjectId("6ab0fe44f620004aca5e8a09"),
  "user_id": ObjectId("6ab0f91bf620004aca5e8a07"),
  "action": "Perform Week 7 Practical",
  "timestamp": ISODate("2026-09-21T09:52:04.478+08:00") }
21_September_2026_at_09:52:04 UTC
__v: 0
```

```
{ "_id": ObjectId("6ab10015f620004aca5e8a0c"),
  "user_id": ObjectId("6ab0f91bf620004aca5e8a07"),
  "action": "SEARCH_PYTHON",
  "timestamp": ISODate("2026-09-21T09:59:49.270+08:00") }
21_September_2026_at_09:59:49 UTC
__v: 0
```

MongoDB Compass - 127.0.0.1:27017/digital_library.users

Connections Edit View Collection Help

Compass

My Queries

Data Modeling

CONNECTIONS (3)

Search connections

- 127.0.0.1:27017
 - BookFlow
 - College
 - admin
 - config
 - digital_library
 - activities
 - users
 - local
 - student_records
- Dbmsl
- Week-3_Practical

127.0.0.1:27017 > digital_library > users

Documents 4 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or Generate query Explain

25 1 - 4 of 4

```
{ "_id": ObjectId("6ab0f735f620004aca5e8a05"),
  "username": "Divya",
  "email": "divya@gmail.com",
  "__v": 0 }
```

```
{ "_id": ObjectId("6ab0f7fef620004aca5e8a06"),
  "username": "Chaitanya",
  "email": "Chaitanya27@gmail.com",
  "__v": 0 }
```

```
{ "_id": ObjectId("6ab0f91bf620004aca5e8a07"),
  "username": "Chaitanya",
  "email": "Chaitanya27@gmail.com",
  "__v": 0 }
```

```
{ "_id": ObjectId("6ab10324f620004aca5e8a0e"),
  "username": "Chaitanya",
  "email": "Chaitanya27@gmail.com",
  "__v": 0 }
```

Folder Structure:

Week 7

```
|__nodememo
  |_user-service
    |
    |__ controllers
    |   |__ activityController.js
    |   |__ userController.js
    |
    |__ routes
    |   |__ activityRoutes.js
    |   |__ userRoutes.js
    |
    |__ app.js
    |__ db.js
    |__ package.json
    |__ .env
```

db.js

```
const mongoose = require("mongoose");

const MONGO_URI =
  process.env.MONGO_URI ||
  "mongodb://127.0.0.1:27017/digital_library";

mongoose.connect(MONGO_URI)
  .then(() => console.log("MongoDB connected"))
  .catch(err => console.error("MongoDB connection error:", err));

const userSchema = new mongoose.Schema({
  username: {
    type: String,
    required: true
  },
  email: {
    type: String,
    required: true
  },
  profile: {
    type: mongoose.Schema.Types.Mixed,
    default: {}
  }
});

const activitySchema = new mongoose.Schema({
  user_id: {
    type: mongoose.Schema.Types.ObjectId,
    required: true,
    ref: "User"
  },
  action: {
    type: String,
    required: true
  },
  timestamp: {
    type: Date,
    default: Date.now
  }
});

const User = mongoose.model("User", userSchema);
const Activity = mongoose.model("Activity", activitySchema);

module.exports = { User, Activity };
```

controllers/userController.js

```
const { User } = require("../db");

const createUser = async (req, res) => {
  try {
    const { username, email, profile } = req.body;

    const user = await User.create({
      username,
      email,
      profile
    });

    res.status(201).json({
      message: "User created successfully",
      user
    });
  } catch (error) {
    res.status(500).json({
      message: "Error creating user",
      error: error.message
    });
  }
};

const getUser = async (req, res) => {
  try {
    const user = await User.findById(req.params.id);

    if (!user)
      return res.status(404).json({
        message: "User not found"
      });

    res.json(user);
  } catch (error) {
    res.status(500).json({
      message: "Error getting user",
      error: error.message
    });
  }
};

module.exports = { createUser, getUser };
```

controllers/activityController.js

```
const mongoose = require("mongoose");
const { Activity } = require("../db");

const createActivity = async (req, res) => {
  try {
    const { user_id, action } = req.body;

    if (!mongoose.Types.ObjectId.isValid(user_id))
      return res.status(400).json({
        message: "Invalid user_id"
      });

    const activity = await Activity.create({
      user_id,
      action
    });

    res.status(201).json({
      message: "Activity created successfully",
      activity
    });
  } catch (error) {
    res.status(500).json({
      message: "Error creating activity",
      error: error.message
    });
  }
};

const getActivities = async (req, res) => {
  try {
    const { user_id } = req.params;

    if (!mongoose.Types.ObjectId.isValid(user_id))
      return res.status(400).json({
        message: "Invalid user_id"
      });

    const activities = await Activity
      .find({ user_id })
      .sort({ timestamp: -1 });

    res.json(activities);
  } catch (error) {
    res.status(500).json({
      message: "Error getting activities",
    });
  }
};
```

```

        error: error.message
    });
}
};

module.exports = { createActivity, getActivities };

```

routes/userRoutes.js

```

const express = require("express");

const router = express.Router();

const {
    createUser,
    getUser
} = require("../controllers/userController");

router.post("/users", createUser);
router.get("/users/:id", getUser);

module.exports = router;

```

routes/activityRoutes.js

```

const express = require("express");

const router = express.Router();

const {
    createActivity,
    getActivities
} = require("../controllers/activityController");

router.post("/activities", createActivity);
router.get("/activities/:user_id", getActivities);

module.exports = router;

```

app.js

```

const express = require("express");

const app = express();

require("./db");

app.use(express.json());

```

```
const activityRoutes = require("./routes/activityRoutes");
const userRoutes = require("./routes/userRoutes");

app.use("/", activityRoutes);
app.use("/", userRoutes);

app.listen(5001, () => {
  console.log("User service running on port 5001");
});
```